

VAHAGN GHAZARYAN

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EDUCATION

Shanghai Jiao Tong University

Sep 2021 – June 2025

B.Sc. in Software Engineering

Courses: Computer Architecture, Operating Systems, Advanced Data Structures, Compilers, Cloud Computing.

WORK EXPERIENCE

Cognaize

Sep 2019 - Sep 2020

Data Engineer Intern

- Collaborated with AI engineers to build ETL pipelines for processing financial documents.
- Developed a unified Python interface for AWS S3, GCS, and Google Drive APIs to standardize storage operations.
- Built REST APIs using Flask to support internal data management workflows.

Institute of Parallel and Distributed Systems

Apr 2024 – Apr 2025

Undergraduate Researcher

- Built tools in Java/Kotlin to benchmark Android applications' performance during LLM agent co-execution.
- Identified bottlenecks and contributed optimizations across the Android framework and Linux kernel layers.

PROJECTS

minik8s: Container Orchestration System

A lightweight Kubernetes-like container orchestration system fully automated for building, testing, and deploying containers.

- Features: Control Plane (API Server, Scheduler, Controllers) and Worker Node (Kubelet, KubeProxy); supports deployment, HPA, DNS, and container lifecycle automation.
- Technologies: Docker, Kubectrl, Etcd, Weave, Iptables, GitHub Actions.
- [GitHub Repository](#)

ClipOnCloud: Confidential Inference Deployment

A privacy-preserving deployment for image-to-video models on AWS EC2.

- Features: Separate execution between Trusted (TEE) and Rich (REE) environments; remote attestation and full input/output encryption.
- Technologies: Flask, torch.distributed, AMD SEV-SNP, Intel SGX.
- [GitHub Repository](#)

Bookstore: Online Shop Platform

A scalable, fault-tolerant, low-latency e-commerce platform.

- Features: Role-based authentication, book search and checkout, order tracking, admin dashboards, and sales analytics.
- Technologies: React.js, Spring Boot, MySQL, MongoDB, Neo4j, Redis, Kafka, Apache Spark, Nginx, and Docker.
- [GitHub Repository](#)

OPEN SOURCE

StringZilla (2.5k+ GitHub stars) - Enhanced CLI support by adding split/wc functionality [\[PR #99\]](#).

SKILLS

Coding: Python, Go, Java, C/C++, JavaScript.

DevOps & Cloud: Docker, Kubernetes, AWS (EC2, S3, RDS, CloudWatch).

Data Engineering: Apache Airflow, Spark, Kafka.

Databases: MySQL, MongoDB, Redis.

CI/CD: GitHub Actions.

Languages: Armenian, English, Russian, Chinese.